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Mini-Project-Synopsis

CUDA Accelerated Background Subtraction for Image Surveillance

SUBMITTED

BY

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INTRODUCTION

1.1 Background and Context

Video surveillance has become an indispensable component of modern security infrastructure, deployed across urban environments, transportation hubs, industrial facilities, and public spaces. At its core, effective surveillance depends on the accurate detection and tracking of moving objects within video streams.

Background subtraction is a classical computer vision technique that separates foreground objects (people, vehicles, anomalies) from a learned static or quasi-static background model. Algorithms such as Gaussian Mixture Models (GMM), frame differencing, and adaptive background modeling have been studied extensively. However, applying these methods to high-resolution, real-time, multi-camera surveillance streams introduces significant computational demands that exceed the capabilities of conventional CPU-based processing pipelines.

Graphics Processing Units (GPUs), originally designed for rendering pipelines, have evolved into massively parallel general-purpose processors capable of executing thousands of threads concurrently. NVIDIA's Compute Unified Device Architecture (CUDA) provides a programming model and API that unlocks this parallelism for scientific and engineering computation. The inherently data-parallel nature of image and video processing where the same operation is applied independently to millions of pixels makes CUDA an ideal accelerator for background subtraction workloads.

This project explores the design and implementation of a CUDA-accelerated background subtraction system targeting real-time surveillance scenarios. By offloading per-pixel computations to the GPU, the system aims to achieve frame processing rates sufficient for live video analysis while maintaining detection accuracy comparable to CPU-based reference implementations.

1.2 Motivation

The proliferation of high-definition and 4K surveillance cameras, combined with the trend toward centralized multi-feed monitoring, has created a bottleneck in real-time video analytics. Traditional CPU implementations process frames sequentially and struggle to meet the latency requirements of live surveillance. GPU acceleration through CUDA presents a direct path to overcoming this bottleneck, enabling analytics pipelines that scale with camera density without proportional increases in hardware cost.

Furthermore, advances in adaptive background modeling such as the ViBe (Visual Background Extractor) and MOG2 (Mixture of Gaussians v2) algorithms present algorithmic complexity that benefits substantially from parallel execution. This project is therefore motivated by the intersection of practical surveillance engineering needs and the theoretical potential of GPU-parallel computer vision.

LITERATURE REVIEW

2.1 Background Subtraction Algorithms

Stauffer and Grimson (1999) introduced the foundational Gaussian Mixture Model (GMM) for adaptive background estimation, modeling each pixel as a mixture of Gaussians whose parameters are updated online. This work demonstrated robust handling of gradual illumination changes and multimodal backgrounds but at significant computational cost per pixel.

Zivkovic (2004) refined this approach with the MOG2 algorithm, introducing automatic selection of the number of Gaussian components per pixel and improving adaptation speed. While more efficient than the original GMM, MOG2 still imposes a per-pixel workload that scales linearly with image resolution.

Barnich and Van Droogenbroeck (2011) proposed ViBe, a non-parametric background modeling approach that maintains a sample-based pixel model and employs spatial propagation of background updates. ViBe achieves fast initialization from a single frame and demonstrates strong detection performance, making it well-suited for GPU implementation due to its regular, data-parallel structure

2.2 GPU and CUDA-Based Acceleration

Grauer-Gray et al. (2009) presented early work on GPU-accelerated background subtraction, demonstrating that mapping pixel-level operations to CUDA threads achieves speedups of 10-20x over CPU implementations for standard definition video. Their work established the viability of the CUDA thread-per-pixel mapping paradigm.

Extended CUDA-based background subtraction to HD content, addressing memory bandwidth constraints through shared memory optimization and warp-level data reuse. They reported near-real-time performance at 1080p resolution with modest GPU hardware.

More recently, Cioppa et al. (2020) explored deep learning-based background subtraction (SuBSENSE and PAWCS) and noted that even neural approaches benefit from GPU parallelism for inference speed, reinforcing the central role of CUDA in modern video analytics pipelines.

OpenCV's CUDA module provides production-grade GPU-accelerated implementations of MOG2 and other background subtractors, serving as a practical baseline for performance comparison in applied projects.

2.3 Research Gaps

While existing literature validates GPU acceleration for background subtraction in controlled settings, several gaps remain: (i) comprehensive benchmarking across diverse surveillance scenarios (indoor, outdoor, night, crowd); (ii) evaluation of custom CUDA kernels versus library-based approaches (OpenCV CUDA) for the same algorithm; and (iii) analysis of the accuracy-throughput trade-off under varying GPU memory constraints. This project addresses these gaps through a structured implementation and evaluation study

PROBLEM STATEMENT

Real-time video surveillance systems are required to process continuous high-resolution video streams at frame rates sufficient for live monitoring, typically 25–30 fps for HD (1080p) content. Background subtraction, a fundamental preprocessing step for foreground detection, involves per-pixel model updates and classification operations across every frame. On a standard CPU, processing a 1920×1080 frame with MOG2 requires approximately 33–50 ms per frame — marginally feasible for single-camera feeds but entirely inadequate for multi-camera deployments where dozens of streams must be processed concurrently.

The core problem addressed by this project is the computational bottleneck introduced by CPU-sequential execution of background subtraction algorithms in high-throughput surveillance pipelines. Specifically:

- CPU-based implementations cannot sustain real-time processing (< 33 ms/frame at 30 fps) for HD resolution video at scale.
- The per-pixel independence of background model update and foreground classification operations is fundamentally underutilized by sequential CPU execution.
- Existing CUDA-accelerated solutions either rely entirely on library abstractions (obscuring optimization opportunities) or lack systematic evaluation against standardized surveillance datasets.

This project therefore aims to design, implement, and rigorously evaluate a CUDA-based background subtraction pipeline that exploits GPU parallelism to achieve real-time throughput on HD surveillance video, while quantifying the accuracy and performance trade-offs relative to CPU baselines.

OBJECTIVES

The specific aims of this project are as follows:

1. Implement a CPU-baseline background subtraction system using the MOG2 or ViBe algorithm in C++/OpenCV to serve as a reference for performance and accuracy measurement.
2. Design and implement custom CUDA kernels for the selected background subtraction algorithm, mapping per-pixel model update and foreground classification operations to individual GPU threads, with optimizations including shared memory usage and coalesced global memory access.
3. Develop a comparative evaluation framework that measures: (a) frame processing throughput (fps) and latency (ms/frame) at 720p and 1080p resolution; (b) GPU memory utilization; and (c) foreground detection accuracy (F-measure, precision, recall) using benchmark datasets such as CDNet 2014.
4. Profile the CUDA implementation using NVIDIA Nsight Systems and Nsight Compute to identify and resolve memory bandwidth bottlenecks, kernel occupancy limitations, and warp divergence issues.
5. Compare custom CUDA kernel performance against OpenCV's built-in CUDA-accelerated MOG2 to quantify the benefit of hand-optimized kernels over library implementations.
6. Document architectural decisions, optimization strategies, and lessons learned to contribute a replicable reference implementation for CUDA-accelerated surveillance analytics.

These objectives collectively target the demonstration that GPU parallelism, when explicitly programmed via CUDA, yields real-time background subtraction performance at HD resolution reducing per-frame latency by a factor of 10x or greater compared to CPU baselines while maintaining statistically comparable detection accuracy

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